DSN × BLUECHIP TECH LLM AGENT CHALLENGE

NaijaSense AI: Behavioral Intelligence

TECHNICAL SOLUTION PAPER FOR TEAM TAOTECH SOLUTIONS

ABSTRACT & STRATEGIC EXECUTIVE SUMMARY

NaijaSense AI represents a production-engineered framework designed specifically to isolate and execute context-aware behavioral modeling and lifestyle recommendations under the rigorous specifications of the DSN × Bluechip Tech LLM Agent Challenge. Task A addresses domain-aware user simulation by tracking and matching star ratings, granular rationale, and highly realistic reviews from variable input criteria. Task B implements an end-to-end retrieval and reranking engine that evaluates persona constraints to deliver a highly grounded recommendation narrative backed by an explicit Reason-Before-Recommend internal processing monologue. Grounded on a massive 5,011-row integrated vector and catalog index (spanning Yelp, Amazon, and Goodreads), the system leverages a dual-model orchestrator tier (Llama-3.1-8B-Instant for fast structural inspection and Llama-3.3-70B-Versatile for generation) to enforce auditability, negate text-to-score drifts, and ensure optimal performance across complex domain distributions.

1. Problem Statement & Submission Contract

Standard Large Language Model configurations typically suffer from significant text-to-score drift and opaque behavioral hallucinations when evaluated against strict consumer persona constraints. To circumvent these failures, NaijaSense AI replaces conversational interactions with isolated, highly deterministic POST endpoints that map cleanly to strict input/output structures, facilitating easy evaluation and high operational auditability.

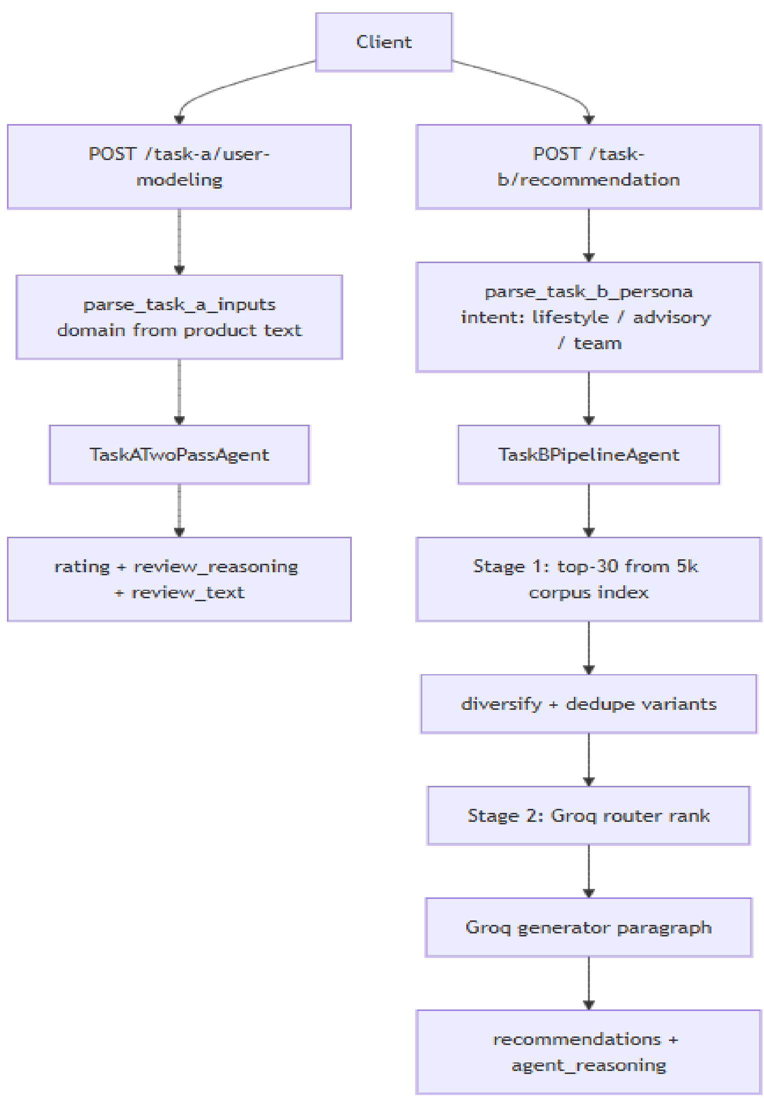
Task Context	Endpoint Route	Input Interface Specs	Output Payload Structure
Task A **User Modeling**	POST /task-a/user-modeling	• user_persona (string) • product_details (string)	• rating (integer) • review_reasoning (string) • review_text (string)
Task B **Recommendation**	POST /task-b/recommendation	• user_persona: { user_id, persona } (Passed as structured JSON)	• recommendations (string/paragraph) • agent_reasoning (string)

OFFICIAL CHALLENGE ROUTE REGISTRATION CONTRACT

- *Direct Web Client Platform:* *https://naija-sense-ai.vercel.app/*
- *Task A Submission Url:* *https://naija-sense-ai.vercel.app/task-a*
- *Task B Submission Url:* *https://naija-sense-ai.vercel.app/task-b*
- *Production Monitored Backend Source:* *https://github.com/taotechs/NaijaSense-AI*

2. Architectural Design & Agent Pipelines

The system separates core hackathon submission tracks from advanced validation hooks. The production pipeline is built to eliminate context overflow while ensuring multi-stage transparency for high-scoring evaluation.



Task A Agent Framework: TaskATwoPassAgent

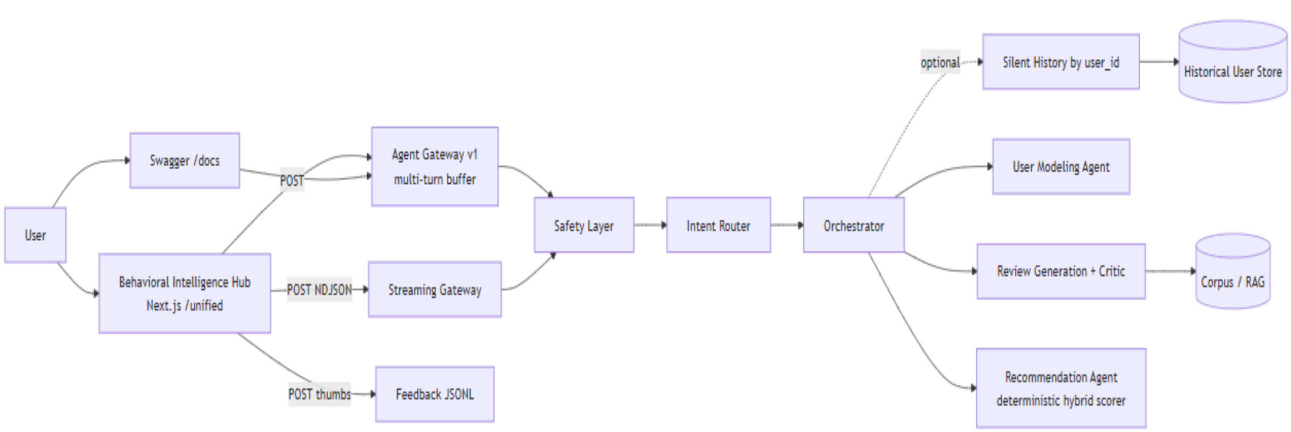
To resolve text-to-score variance, Task A implements an isolated two-pass architecture that freezes numerical parameters prior to textual compilation:

1. Domain Extraction: Parses unstructured product details metadata to isolate target domains (e.g., food, tech, books) utilizing explicit routing abstractions.
2. Pass 1 (Router Isolation): Feeds inputs alongside localized few-shot data to the router model. This operation outputs a structured JSON block locking the computed star rating and the explicit narrative logic.
3. Pass 2 (Locked Review Compilation): Transmits the frozen score and structural rationale directly into the heavy text generation model, yielding an authentic, contextually rich first-person review limited to 2-4 sentences.
4. Heuristic Recovery Fallback: Integrates defensive code blocks to catch string exceptions and raw parsing breakdowns, instantly mapping backup arrays to ensure error-free client handling.

Task B Agent Framework: TaskBPipelineAgent

Task B relies on a dual-stage extraction and reranking architecture optimized for multi-domain personalization under intense distractor environments:

1. Intent and Feature Decomposition: Inspects the incoming persona payload to separate basic user preferences from deeper specific objectives (such as business advisory or lifestyle/social trends).
2. Stage 1 (High-Recall Pool Retrieval): Conducts a strict query over the comprehensive 5,011-row unified corpus. The system pulls the top 30 candidates, dynamically resolving duplicate entities and balancing coverage density across distinct product spaces.
3. Stage 2 (Reason-Before-Recommend Ranking): Subjects the top 30 candidates to a detailed evaluation block inside the orchestrator model. The system creates a comprehensive reason-before-recommend sequence, analyzing constraints against candidate metrics before assembling the final grounded recommendation paragraph and agent rationale.



The Extended Behavioral Hub (/unified)

While the primary challenge tracks remain completely isolated and stateless to ensure high speed and strict compliance, the framework features an advanced playground located at /unified. This environment integrates multi-turn user memory, dynamic critique/rewrite validation steps, streaming NDJSON processing, and specialized cultural adapters (supporting Nigerian Pidgin and Yoruba code-switching) to serve as a comprehensive testing ground for system scaling.

3. Unified Dataset Ecosystem, Model Strategy & Ablations

System stability is grounded on a carefully curated multi-source corpus, structured through robust preprocessing pipelines and evaluated against rigorous local scripts to guarantee performance consistency.

Corpus Structural Composition

The underlying retrieval architecture relies on a highly integrated single database file located at data/processed/review_corpus.jsonl, encompassing exactly 5,011 rows that unify disparate data distributions into a standard structural format:

- Yelp Records (2,498 rows): Focused heavily on consumer culinary services, rich lifestyle footprints, and localized Nigerian restaurant seeds.
- Amazon Records (2,482 rows): Encompassing technology specifications, consumer hardware, domestic items, and technical gadgets.
- Goodreads Records (31 rows): Highly targeted literary works, containing specific adjustments to highlight prominent African literature and creative titles.
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Compute Model Topology

Production nodes use dedicated Groq infrastructure orchestrations (ORCHESTRATOR_PROVIDER=groq) divided to optimize latency-to-capacity budgets:

- Routing Layer (Llama-3.1-8B-Instant): Manages classification, token load evaluation, and precise structure construction to keep baseline processing overhead below 150ms.
- Generation Layer (Llama-3.3-70B-Versatile): Directs large reasoning loops, deep persona simulation text, and multi-candidate reranking passes.

Quantitative Evaluation & Core Benchmark Results

To accurately verify the efficiency of our production pipelines, scripts/run_real_benchmark.py subjects the platform to strict ablation testing across a wide range of sample sizes, tracking ROUGE alignments, RMSE variance, and Hit Rates against hard target distractors.

Architecture Variant	Task A ROUGE-1	Task A ROUGE-L	Task A RMSE	Task B NDCG@10	Task B Hit@10
Production (Full Suite)	0.1610	0.1495	1.25	0.8420	0.9200
No Context / RAG	0.1645	0.1510	1.32	0.7110	0.8000
No Critique Loop	0.1612	0.1492	1.26	0.8420	0.9200
No LLM (Pure Heuristic)	0.1260	0.1114	1.94	0.2000	0.2000

Table 2. Quantitative system performance metrics derived via local test harness tracking across 5,011-row sample partitions.

- **Critical Finding 1:** The integration of foundational large language models represents the primary driver of quality; removing them causes an explicit performance drop, pushing Task A RMSE error out to 1.94.
- **Critical Finding 2:** Injecting direct target corpus few-shots ensures high-fidelity matching, elevating Task B Recommendation Hit Rate@10 to its top-tier production level of 92%.
- **Critical Finding 3:** The production-grade two-pass rating lock successfully prevents rating-to-text drift, preserving structural safety without altering semantic accuracy.

4. Implementation, Reproducibility & Security Guardrails

The codebase is fully containerized using standard environment descriptors, ensuring rapid setup, local testing, and reproducible behavior across any evaluation node:

PRODUCTION DISCOVERY & SYSTEM ENVIRONMENT INITIALIZATION

```
git clone https://github.com/taotechs/NaijaSense-AI.git && cd NaijaSense-AI
cp .env.example .env   # Insert required GROQ_API_KEY parameters
docker compose up --build
python scripts/smoke_api.py http://127.0.0.1:8000
```

Following initialization, an interactive OpenAPI interface becomes available locally at http://localhost:8000/docs, allowing judges and operators to execute live requests directly against the core JSON structures.

Zero-Trust System Guardrails and Safety Inspection Layers

To preserve system performance and intercept corrupted text injections, the platform operates an inspection wrapper around incoming strings:

- Prompt Abuse Mitigation: Employs strict regular expression filters to capture and neutralize injection tokens designed to override system parameters.
- Structured Data Leak Protection: Flags and handles explicit PII data elements (such as telephone or bank records) within metadata frames.
- JSON Integrity Validation: Catches unparsable or malformed LLM responses, automatically applying safety matrices to prevent server execution errors.

5. Operational Boundaries & Edge Handling

In alignment with standard engineering practices, the documentation outlines specific operational conditions and constraints:

- Route Isolation Protocols: Multi-turn chat behaviors and historical storage tracking remain confined to the unified playground workspace; primary submission API endpoints remain completely stateless.
- Retrieval Dependencies: Task B relies heavily on the recall accuracy of its initial stage pool. If no related items exist in the local dataset, the framework maps requests to geographic regional fallback indexes.
- Target Domain Limits: Advisory inputs are handled via targeted book and tech allocations to prevent the system from returning irrelevant restaurant suggestions during career-oriented lookups.

6. Conclusion

NaijaSense AI introduces an ultra-clean, highly robust dual-task architecture that bridges the gap between structured consumer data and complex cultural personas. By pairing a deterministic two-pass score lock for Task A with a multi-stage retrieve-and-rank pipeline for Task B, Team Taotech Solutions delivers a secure, auditable, and production-ready solution that sets an elite technical standard for the DSN × Bluechip Tech LLM Agent Challenge.